

PHYS 427: Lecture # 3

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Lecture Span

1. Thermal equilibrium and temperature

Review

For a paramagnet, σ is a closed parabola with a max at $N = 0$ and is zero at the locations where the spins are all aligned (down and up).

Also, Einstein solid was $\Omega = U^N \implies \sigma = \ln \Omega$. These are all closed systems (zero interaction with surroundings), i.e. U and N are constant.

2 coupled systems

Now, consider 2 coupled systems that can exchange energy but not particles (thermal contact). Consider two systems $A : U_A, N_A$ and $B : U_B, N_B$. If each system is closed, then all variables are constant.

Then

$$\Omega(U_A, U_B) = \Omega(U_A)\Omega(U_B) \quad (1)$$

Now, consider the case that they are in thermal contact. Then, $U = U_A + U_B$ is fixed, but U_A and U_B can vary. N_A and N_B are fixed. Then,

$$\Omega_{\text{tot}} = \sum \Omega_A(U_A)\Omega_B(U - U_A) \quad (2)$$

2 Einstein solids

$\Omega(U) \sim U^N$. Then, plotting

$$\Omega_A(U_A)\Omega_B(U - U_A) \quad (3)$$

You obtain some sort of peak, with a width of the peak as σ_{U_A} (not entropy). The peak in Ω represents the most likely state. Now, find $\max(\ln(\Omega))$ for an Einstein solid. Calculate:

$$\ln(\Omega_A(U_A)\Omega_B(U - U_A)) = \ln(\Omega_A(U_A)) + \ln(\Omega_B(U - U_A)) \quad (4)$$

$$= \partial_{U_A} \ln(\Omega_A(U_A)) + \partial_{U_A} \ln(\Omega_B(U - U_A)) \quad (5)$$

$$\frac{N_A}{\overline{U_A}} - \frac{N_B}{\overline{U - U_A}} = 0 \quad (6)$$

$$\frac{U}{\overline{U_A}} - 1 = \frac{N_B}{N_A} \quad (7)$$

$$\overline{U_A} = \frac{U}{1 + N_B/N_A} \quad (8)$$

$$\overline{U_B} = U - U_A = \frac{U}{1 + N_B/N_A} \quad (9)$$

$$\frac{\overline{U_A}}{N_A} = \frac{\overline{U_B}}{N_B} = \frac{U}{N_A + N_B} \quad (10)$$

In the most likely macrostate, the average energy per oscillator across both systems is equal. The width of the peak can be calculated as follows. Let $U_A = \overline{U}_A + \delta U$. We calculated previously that

$$\Omega(U) = \left(\frac{eU}{N\hbar\omega} \right)^N \quad (11)$$

Here, we let $U = U_A + \delta U$.

Then plug the above equation into $\ln(\Omega_A) + \ln(\Omega_B)$, and you obtain

$$= 2N + N \ln \left(\frac{\overline{U}_A + \delta U}{N\hbar\omega} \right) + N \ln \left(\frac{\overline{U}_A - \delta U}{N\hbar\omega} \right) \quad (12)$$

$$= 2N - 2N \ln(N\hbar\omega) + N \ln(\overline{U}_A^2 - \delta^2) \quad (13)$$

$$U_A^2 - \delta U^2 = U_A^2 \left(1 - \frac{\delta U^2}{U_A^2} \right) \quad (14)$$

$$= 2N \ln \left(\frac{e\overline{U}_A}{N\hbar\omega} \right) + N \ln \left(1 - \frac{\delta U^2}{U_A^2} \right) \quad (15)$$

$$\approx 2N \ln \left(\frac{eU_A}{N\hbar\omega} \right) - \frac{N\delta U^2}{U_A^2} \quad (16)$$

Finally, exponentiating both sides yields

$$\Omega_A \cdot \Omega_B = \left(\frac{eU_A}{N\hbar\omega} \right)^{2N} \exp \left(-\frac{N\delta U^2}{U_A^2} \right) \quad (17)$$

This means that, up to a Taylor expansion around U_A , this curve is a gaussian. Then, we find that (the standard deviation)

$$\sigma_{U_A} = \frac{U_A}{\sqrt{2N}} \quad (18)$$

$$\frac{\sigma_{U_A}}{U_A} = \frac{1}{\sqrt{2N}} \quad (19)$$

Therefore, for $N \gg 1$, we have that this relative width goes to 0. Then, we can conclude that

$$\Omega_{\text{tot}} = \sum_{U_A} \Omega_A \Omega_B \approx \Omega_A(\overline{U}_A) \Omega_B(U - \overline{U}_A) \quad (20)$$

Next, consider entropy

$$\sigma = \ln \Omega \quad (21)$$

Now, bring two systems into thermal contact at $t = 0$. Then,

$$\sigma_{\text{tot}} = \sigma_A(U_A) + \sigma_B(U - U_A) \quad (22)$$

We assume initially that $U_A(t = 0) > U_B(t = 0)$. Initially, $\sigma(t = 0) = \sigma(U_A(t = 0)) + \sigma(U_B(t = 0))$. Then, it evolves into

$$\sigma = \sigma_A(\overline{U}_A) + \sigma_B(U - \overline{U}_A) \quad (23)$$

The system evolves into a configuration that maximizes its total entropy, but entropy of individual systems may go down. This is the 2nd law of thermodynamics. Also,

$$\frac{\partial \sigma_{\text{tot}}}{\partial U_A} = 0 = \partial_{U_A} \sigma_A + \partial_{U_B} \sigma_B \left(\frac{\partial U_B}{\partial U_A} \iff -1 \right) \quad (24)$$

Therefore,

$$\partial_{U_A} \sigma_A = \partial_{U_B} \sigma_B \quad (25)$$

question: maximize σ as a function of U_A ? Or is σ natural log of the total multiplicity (i.e., summing over all multiplicities wrt U_A)

Formal definition of fundamental temperature

$$\frac{1}{\tau} = \left(\frac{\partial \sigma}{\partial U} \right)_N \quad (26)$$

Equilibrium condition states that $\tau_A = \tau_B$ (or the fundamental temperatures are the same).

0-th law of thermodynamics: If A is in thermal equilibrium with systems B and C , then the temps of A, B, C are all the same.

Absolute temperature T measured in Kelvin is

$$\tau = k_B T$$

k_B is the Boltzmann constant. Then, the convention definition of entropy is

$$S = k_B \ln(\Omega) \quad (27)$$

Then,

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_N \quad (28)$$

Also, calculate

$$\Delta S = \Delta S_A + \Delta S_B \quad (29)$$

$$= \frac{\partial S_A}{\partial U_A} \Delta U_A + \frac{\partial S_B}{\partial U_B} \Delta U_B \quad (30)$$

$$= \frac{1}{T_A} \Delta U_A + \frac{1}{T_B} \Delta U_B \quad (31)$$

$$= \left(\frac{1}{T_A} - \frac{1}{T_B} \right) \Delta U_A \geq 0 \quad (32)$$

$$(33)$$

If $T_A > T_B$, then energy must flow from U_A to U_B , i.e., $\Delta U_A < 0$. Therefore, energy flows from hot to cold.

Also, $\Delta U_A = T_A \Delta S_A = -T_B \Delta S_B = -\Delta U_B$. This form of energy transfer is also called heat. As an example, $\Omega \sim U^N$. Then,

$$S = N k_B \ln(U) + f(\text{other variables besides } U) \quad (34)$$

Now,

$$\frac{1}{T} = \frac{\partial S}{\partial U} = N k_B \frac{\partial}{\partial U} \ln U \quad (35)$$

$$U = N k_B T \quad (36)$$